

Bryce Ilcan

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EDUCATION

Schulich School of Engineering, University of Calgary

2024 - Present

Bachelor of Science, Electrical Engineering

GPA: 3.5/4.00

Altium Student Ambassador

September 2026 - Present

SKILLS

Hardware & Power Electronics

PCB schematic capture/layout, DC-DC converters, planar magnetics, soldering/reflow, prototyping, Li-ion pack fabrication, oscilloscope, multimeter

Programming & Firmware

C, C++, Python, JavaScript/TypeScript, Zephyr RTOS, React, FastAPI, PyMAVLink, WebSockets, SystemVerilog

Interfaces & Platforms

I2C, SPI, UART, QSPI, SWD, USB, nRF52840, Raspberry Pi, ArduPilot

Tools

Altium Designer, PLECS, Ansys Maxwell, FEMM, LTspice, Quartus, Fusion 360, Git, Linux, Mission Planner

WORK EXPERIENCE

University of Calgary

May 2026 - Present

Power Electronics Researcher

Calgary, AB

- Developing a 725 W, nominal 324 V-to-12 V phase-shifted full-bridge DC-DC converter with synchronous rectification and a stackable multilevel secondary architecture to reduce output ripple and current per channel
- Calculated planar transformer dimensions and winding parameters; validated magnetic performance in FEMM and Ansys Maxwell and simulated converter operation in PLECS.
- Designed a 4-layer PCB with planar transformer and inductor structures with signal routing and high-power copper planes

Schulich Unmanned Aerial Vehicles (SUAV)

October 2024 - Present

Co-Avionics Lead

Calgary, AB

Junior Executive, July 2025 - February 2026; Avionics Member, October 2024 - June 2025

- Lead electrical development for a 20+ person electrical team within a 60+ member club, coordinating PCB design and aircraft electrical integration.
- Lead custom ESC development and oversee delegated GPS module, smart DC-DC converter, and instrumented motor test-stand projects, providing technical guidance to project leads
- Integrated a Septentrio mosaic GNSS receiver into the aircraft avionics system, supporting navigation and overall electrical integration
- Led a three-person team creating a BQ76952-based Li-ion monitoring PCB in Altium with multiplexed I2C communication to a Raspberry Pi
- Developed C++ plugins for Lockheed Martin VCSI and supported autonomous mission planning, helping the team place first out of 81 teams in SUAS 2025 design deliverables and ninth overall at C-UASC 2026

ENGINEERING PROJECTS

Custom nRF52840 Bike Computer

May 2026 - Present

- Designed a custom 4-layer nRF52840-based PCB in Altium Designer, integrating GNSS over UART, an accelerometer and barometer over I2C, QSPI flash memory, an SPI LCD, USB/SWD programming, and ESD protection
- Created a 1-cell LiPo power subsystem combining four functions: BQ24074 power-path charging, battery protection, fuel gauging, and automatic USB/battery source selection

STM32 Bluetooth-Controlled RC Car

January 2026 - April 2026

- Authored C firmware integrating Bluetooth/UART commands, PWM motor and servo control, LEDs, and ultrasonic sensing; verified operation on hardware and added UART message-alignment checks

Garmin Embedded in Embedded Program

October 2025 - March 2026

- Completed a Garmin engineer-led embedded systems program, writing C code during guided lessons using Zephyr RTOS on nRF52840 hardware

Autonomous Drone and Web Ground-Control Station

May 2025 - August 2025

- Assembled and configured an ArduCopter-based drone that achieved autonomous flight; developed a custom web ground-control station and tested vehicle commands in software-in-the-loop simulation
- Fabricated 6S1P, 4,200 mAh Li-ion battery packs through spot welding, soldering, and wrapping